AP PRECALCULUS

Unit 1 • Worksheet 1.11A

Worksheet 1.11A • Rational Functions and Concept of Limit

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 3} \dfrac{x^2 - 3x}{x - 3}$ is

(A) 0

(B) 1

(C) 9

(D) 3

Question 2

NO CALCULATOR

$\lim_{x \to 2} \dfrac{x^2 - x - 2}{x^2 - 4}$ is

(A) $\frac{3}{4}$

(B) 0

(C) $\frac{1}{2}$

(D) It does not exist.

Question 3

NO CALCULATOR

$\lim_{x \to -1^-} \dfrac{x + 3}{x + 1}$ is

(A) ∞

(B) $-\infty$

(C) 2

(D) 1

Question 4 NO CALCULATOR

Which describes $\lim_{x \to 4} \frac{1}{x-4}$?

(A) ∞
(B) $-\infty$
(C) 0
(D) It does not exist (the one-sided limits are $-\infty$ and ∞).

Question 5 (Free response) NO CALCULATOR

Let $r(x) = \frac{2x^2+2x-4}{x^2-1}$. Describe each feature with a limit statement: (a) the hole, (b) the vertical asymptote, (c) the end behavior.

Question 6 (Free response) NO CALCULATOR

Find each limit, or explain why it does not exist. (a) $\lim_{x \to -2} \frac{x^2+5x+6}{x+2}$ (b) $\lim_{x \to 0^+} \frac{x-1}{x^2}$ (c) $\lim_{x \to \infty} \frac{3-x^2}{2x^2+x}$ (d) $\lim_{x \to 1} \frac{x}{x-1}$